# Megha Pranay Raparla

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## CAREER OBJECTIVE

Computer Science undergraduate (graduating 2028) who builds computer-vision, automation and backend systems, with a software engineering internship and 19 bug fixes merged into OpenCV and CCExtractor. Looking for software engineering and research internships for Summer 2027.

## EDUCATION

**B.Tech in Computer Science and Engineering** 2024 – 2028
Amrita Vishwa Vidyapeetham, Coimbatore CGPA: 8.34 / 10
**Subjects:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## INTERNSHIP EXPERIENCE

**Software Engineer Intern — Nordic Tech Design** Apr 2026 – Jun 2026
Gurgaon, India (On-site) · Sole developer of the core software in a 5-person team

**Purpose:** Automate quality inspection of industrial bolts with a camera, replacing slow manual checks by operators.
**Tech Stack:** Python, OpenCV
**Results:** Measured bolt dimensions to within 2 µm across 40 repeated runs, on par with commercial inspection systems; the defect checker caught a surface pit the human operator had missed.
**Use Cases:** Measuring thread pitch, diameters and hole positions; finding surface defects by comparing each part with a perfect reference part; counting cross-drilled holes from video of a rotating bolt.
**Portability & Scalability:** Built for bolt inspection, but the measuring and defect-checking steps are general and can be scaled to inspect other machined parts.

## PROJECTS

**SmartStore ERP — Store & Inventory Management System** Java, Spring Boot, MySQL
*github.com/pranayr710/SmartStore-ERP*

- Java backend for inventory, orders and suppliers: 50 REST endpoints over an 8-table MySQL database.
- Role-based login and dashboards for admin, staff and customer; reorder alerts; reports exported to Excel and PDF.

**Lumina — Visual AI Workflow Automation Builder** TypeScript, Next.js, Python, FastAPI
*github.com/pranayr710/lumina*

- Drag-and-drop builder for automated workflows: camera or mic input → AI → alerts by email, Slack or Sheets.
- 33 ready-made blocks; describe a workflow in plain English and it builds it; exports workflows to Node-RED.

**ClassGraph — Classroom Engagement Analytics** Python, FastAPI, PyTorch
*github.com/pranayr710/cv*

- Turns a classroom camera feed into a live dashboard of each student's attention, with a report per session.
- Backed by 743 automated tests, so changes are checked before they are relied on.

## OPEN SOURCE CONTRIBUTIONS

**OpenCV** · image-processing library · 11 pull requests merged Sep 2026 – Present

- Fixed cv::sum(), convertTo() and others giving wrong results on large values; every fix passed code review and tests.

**CCExtractor** · C tool that extracts subtitles from video · 8 pull requests merged Sep 2026 – Present

- Fixed dropped video frames, out-of-bounds memory reads, a crash, a memory leak and a Windows-only bug.

## TECHNICAL SKILLS

- **Languages:** Java, Python, SQL, C, JavaScript, TypeScript
- **Backend & Databases:** Spring Boot, REST APIs, FastAPI, MySQL, SQLite
- **Software Engineering:** SDLC, object-oriented design, Git & GitHub, code review, automated testing, debugging
- **Web & Data:** React, Next.js, Power BI, Pandas, OpenCV
- **Tools:** Linux, Docker, Maven, IntelliJ, VS Code, Postman

## ACHIEVEMENTS & ACTIVITIES

- **Google Student Ambassador,** Amrita Vishwa Vidyapeetham
- **6th place, Under-11 National Chess Championship** (2016) · Chess rating: 1406
- **Vice-Captain,** Chess Club, Amrita · **Captain,** CSE Chess Team
- **Member,** Amrita Center for Entrepreneurship (ACE)